

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 5

Total Marks:25

Annexure A

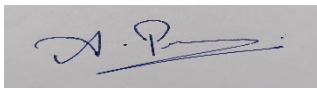
Concept Mapping

Code of Conduct:

I PRASANNA A (192525130) (**Name / Reg No**) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Concept Mapping

Question 1 (5 Marks)

Construct a multi-level concept map integrating:

Operating System Types → OS Structures → System Calls → OS Functions.

Your map must show cross-links (e.g., how OS structure influences system call implementation).

Question 2 (5 Marks)

Design a comprehensive concept map representing:

Process Management, Memory Management, File System, and I/O System.

Include interdependencies and indicate where system calls interact with each component.

Question 3 (5 Marks)

Create a comparative concept map for:

Batch OS, Time-Sharing OS, Distributed OS, and Real-Time OS.

Highlight differences, similarities, advantages, and limitations using linking phrases.

Question 4 (5 Marks)

Develop a dynamic concept map showing the execution flow:

User Program → API → System Call Interface → Kernel → Hardware → Response back to User.

Include feedback loops and control flow paths.

Question 5 (5 Marks)

Construct a hierarchical + functional concept map for system calls:

Process, File, Device, and Information system calls.

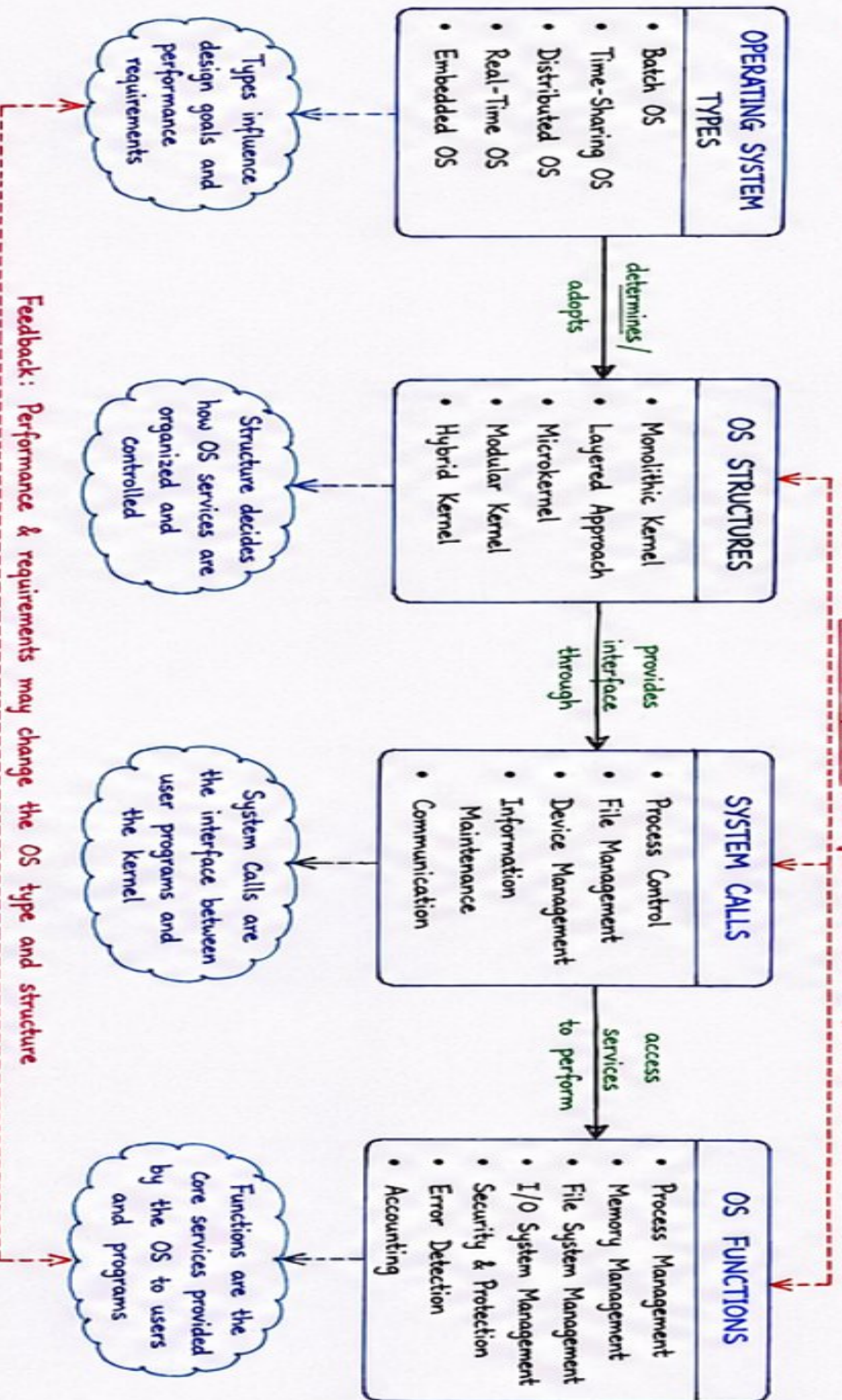
Also show how they relate to OS services like scheduling, memory allocation, and device handling.

Q1. Construct a multi-level concept map integrating:

Operating System Types → OS Structures → System Calls → OS Functions.

(Show cross-links)

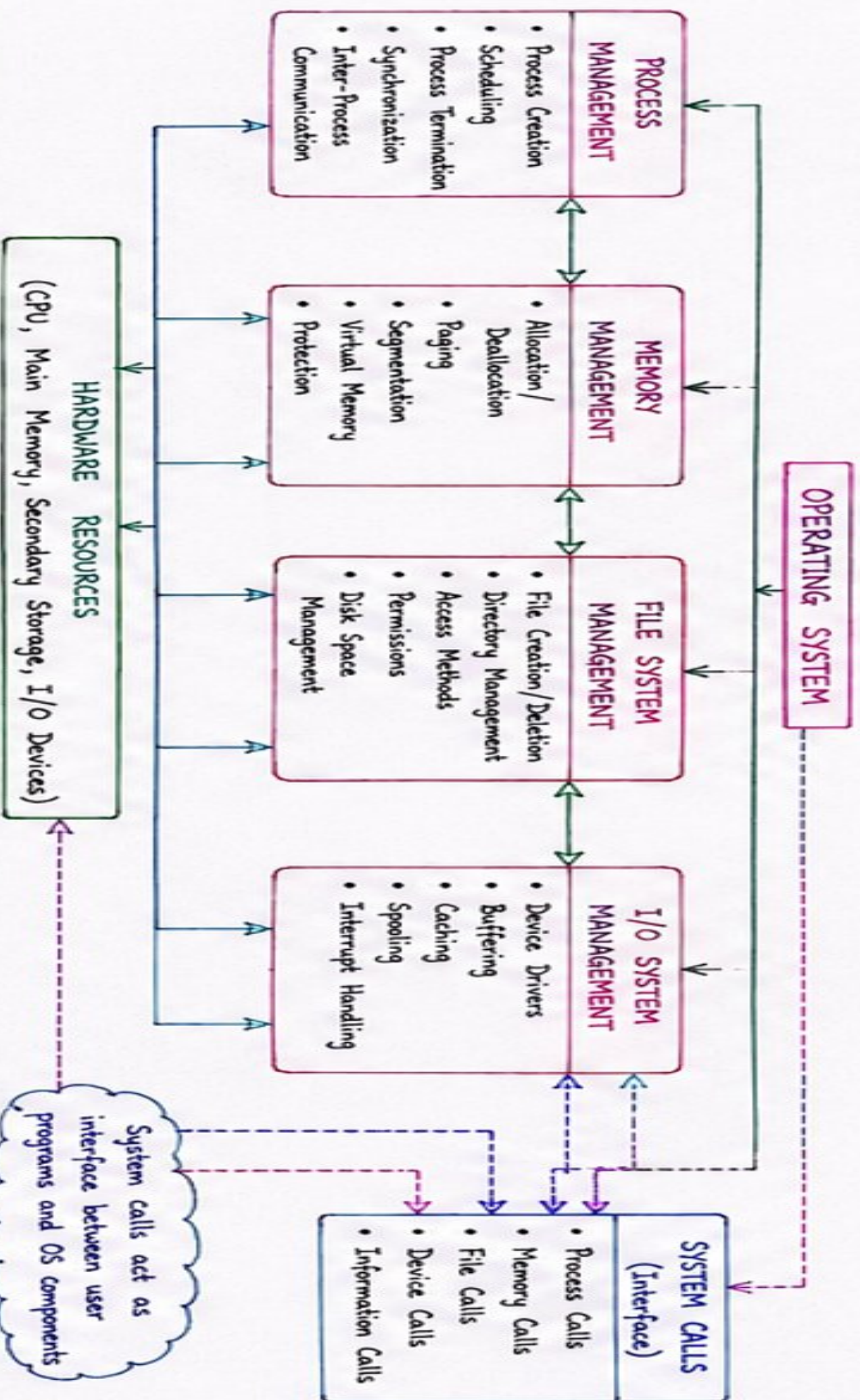
Structure influences how system calls are implemented



Q2. Design a comprehensive concept map representing:

Process Management, Memory Management, File System, and I/O System.

Include interdependencies and indicate where system calls interact.



Q3. Create a comparative concept map for:

Batch OS, Time-Sharing OS, Distributed OS, and Real-Time OS.

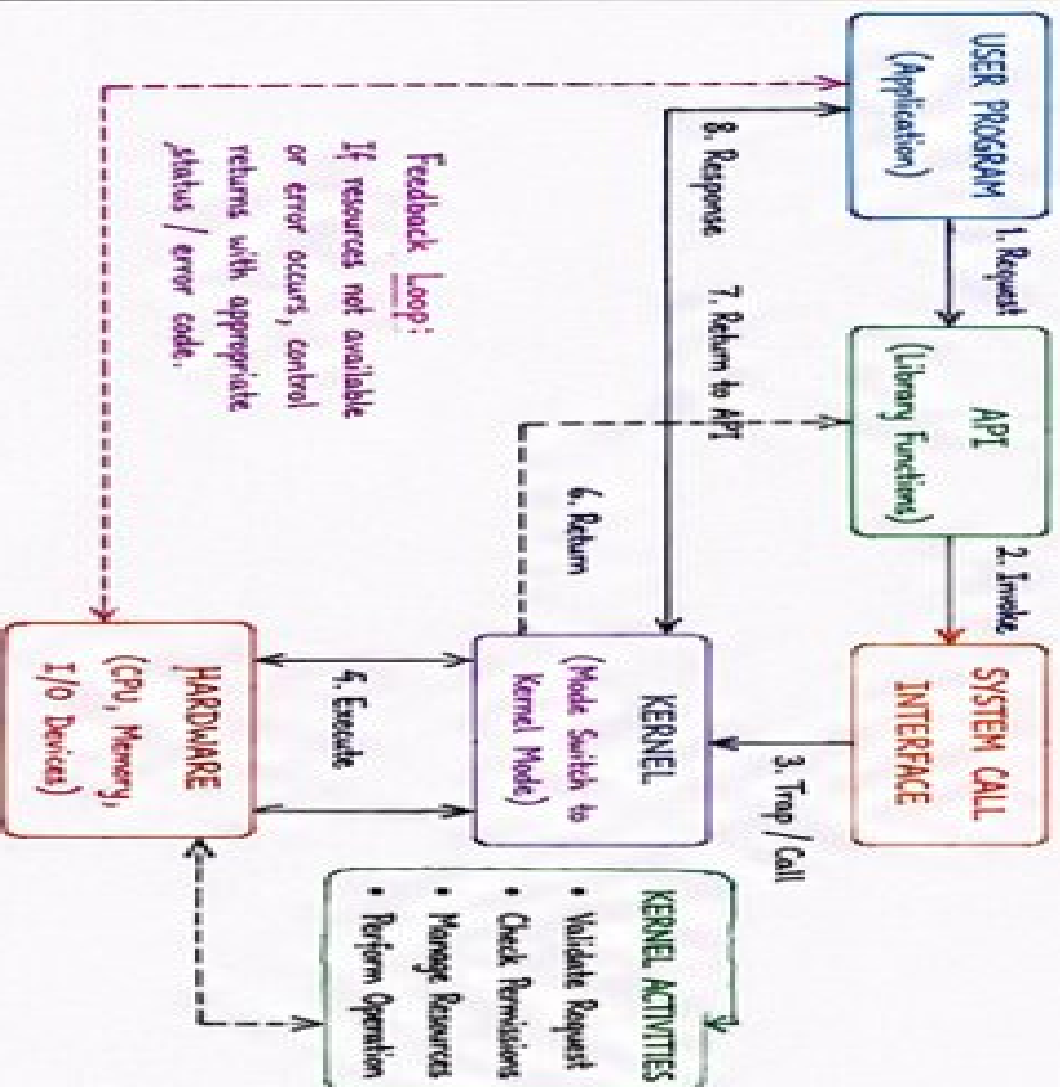
Highlight differences, similarities, advantages, and limitations.

	BATCH OS	TIME-SHARING OS	DISTRIBUTED OS	REAL-TIME OS
Definition	Executes jobs in batches without user interaction.	Allows multiple users to interact with system simultaneously.	Manages multiple computers and makes them appear as one.	Provides guaranteed response within strict time constraints.
Key Features	- Long job queues - High throughput - No interaction	- Multi-user - Short response time - CPU time sharing	- Resource sharing - Scalability - Fault tolerance	- Deterministic - Priority-based - Timely response
Advantages	- Good for large jobs - High CPU utilization	- Quick response - Better user satisfaction	- High performance - Resource availability	- Reliability - Used in critical systems
Limitations	- Slow response time - Difficult debugging	- Complex to manage - Security concerns	- Complex communication - High cost	- Complex design - Limited flexibility
Examples	- IBM OS/360 - Early mainframe OS	- UNIX - Linux, Windows	- Amoeba - Plan 9, Sprite	- Vxworks - QNX, RTLinux

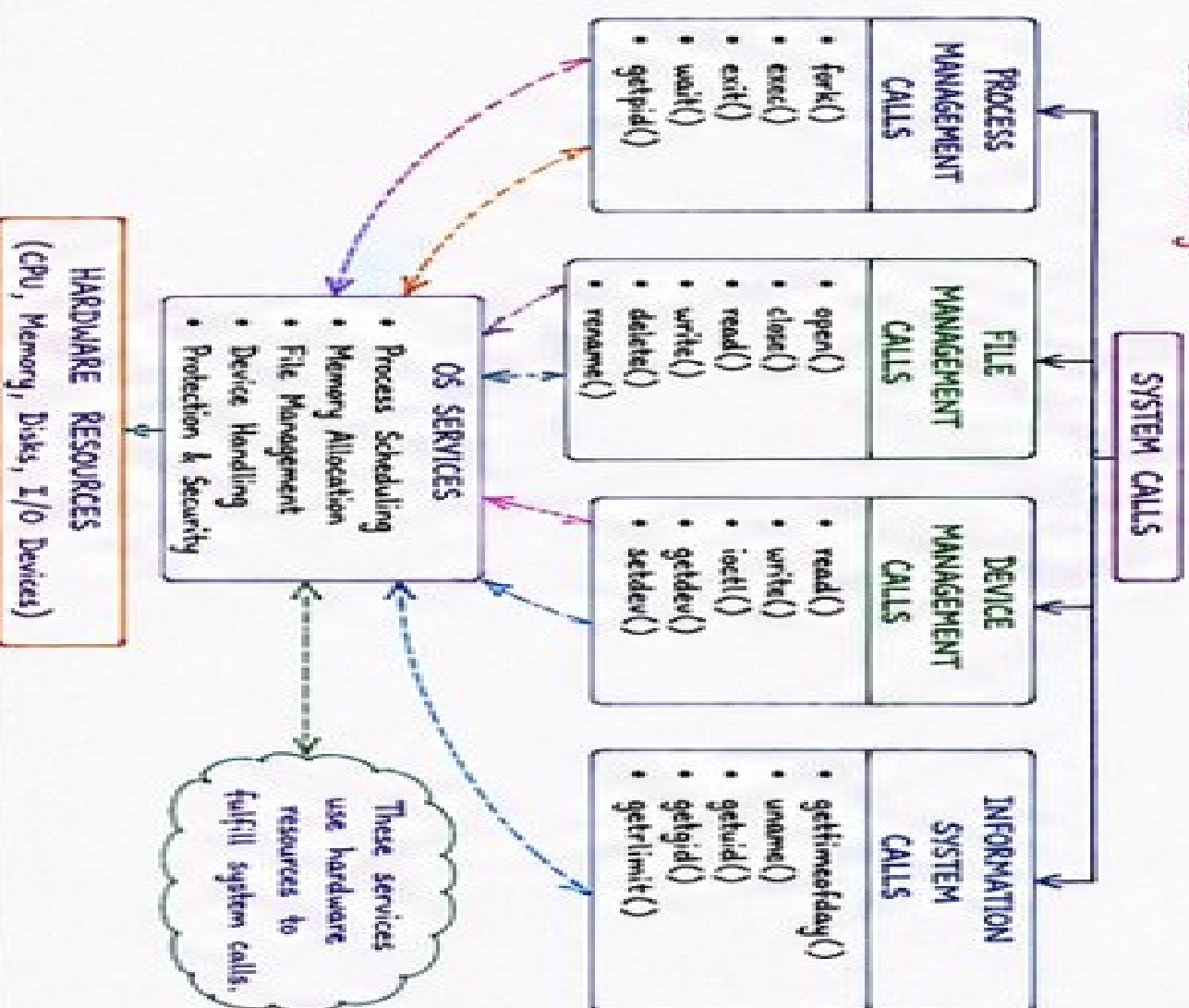
Similarities

- Similarities
- All are types of Operating Systems.
 - Manage resources.
 - Provide system calls.
 - Ensure program execution.

Q4. Develop a dynamic concept map showing the execution flow:
 User Program → API → System Call Interface → Kernel →
 Hardware → Response back to User.
 Include feedback loops and control flow paths.



Q5. Construct a hierarchical + functional concept map for system Calls:
 Process, File, Device, and Information system calls. Also show how
 they relate to OS services like scheduling, memory allocation, and
 device handling.



RUBRICS FOR CALCULATION :

Concept Mapping					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand